

ENIAC

The machines that changed how we think.

ENIAC takes its name from the 30-ton machine that helped begin the electronic computing age. From that room-sized giant to the intelligence in your pocket, we report on one continuous story — the machines we make, and what they make of us.

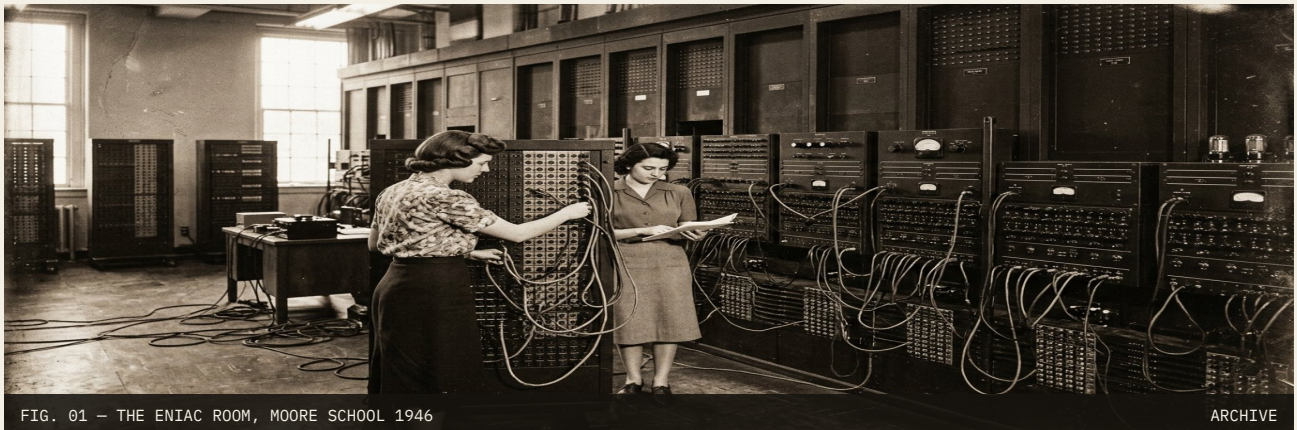


FIG. 01 — THE ENIAC ROOM, MOORE SCHOOL 1946

ARCHIVE

ENIAC is a digital magazine about computing, artificial intelligence, cybersecurity, digital life and the human relationship with technology. This is Issue 01 — THE FIRST ISSUE, published SEPTEMBER 2026. It contains six stories following one thread: from the first counting machines to the intelligence now in our pockets. The machine changes. The human question remains.

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FROM THE EDITOR

Every tool we have ever built was a bet on human curiosity. The abacus. The press. The vacuum tube. The neural network. This issue travels from a machine that weighed thirty tons to the artificial intelligence in your pocket — and asks the only question that matters: what are we for, now that machines can think?

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The machine changes. The human question remains.

*Every tool we have ever built was a bet on human curiosity.
From the abacus to the neural network.*

Every tool we have ever built was a bet on human curiosity. The abacus. The press. The vacuum tube. The neural network.

This issue travels from a machine that weighed thirty tons to the artificial intelligence in your pocket — and asks the only question that matters: what are we for, now that machines can think?

The ENIAC room in 1946 was a wall of black panels, cables and switches, with two figures working at the machine. It weighed 30 tons, contained 17,468 vacuum tubes, drew 150 kilowatts, and could do 5,000 additions per second. Extraordinary for its era. Primitive by ours. Yet its idea never switched off.

We began with counting. We arrived at creating. Fourteen milestones, five thousand years, one direction of travel. The timeline in this issue gives the early milestones room to breathe and lets the recent ones accelerate — because that is what happened.

ENIAC shipped with no manual and no programming language. Six women mathematicians — Kay McNulty, Betty Jennings, Betty Holberton, Marlyn Wescoff, Fran Bilas, Ruth Lichterman — recruited from the wartime corps of human “computers,” studied its circuits and invented the craft of programming it. For decades their work went largely uncredited. History has since corrected the record.

From that giant, we follow three runways that open from a single degree: the Bachelor of Computer Applications. A degree gives you knowledge. BCA gives you the runway to build with it — career, higher studies, creator. You learn to code, build, solve, think. Logic, patience, teamwork, adaptability.

In the physical world we don’t open the door to strangers. Online, we sometimes do it with one click. Cybersecurity is not just clever code; it is human emotion — fear, greed, curiosity — exploited at scale. Five golden rules close that door: stop, think, click; protect your OTP; question “free”; use two locks; update.

Our parents grew up with Google. We are growing up with AI. Super-tutor, super-creator, accelerator, new careers — every superpower has a price. Dependence, comparison, deception. Are we using AI, or is AI using us? Five skills cannot be downloaded: critical thinking, creativity, communication, empathy, leadership.

You think you control the game. But every reward, sound and victory is designed to keep your brain engaged. Play → Reward → Dopamine → Repeat. The loop closes itself. The game is not the enemy. Losing control is.

The machine is the subject. Human curiosity is the story. — THE EDITORS, ISSUE 01

From Computer to Artificial Intelligence.

A journey from counting to creating — fourteen milestones, one direction of travel.

We began with the abacus. We arrived at artificial intelligence. For thousands of years, humans have built tools to make thinking faster, easier and more powerful. Early milestones get the room to themselves; the recent ones arrive faster, because that is what happened.

CH. 01 — THE JOURNEY / 5,000 YEARS

c. 3000 BC — ABACUS

Counting becomes a machine — the first revolution in calculation.

1642 — PASCAL'S CALCULATOR

Blaise Pascal builds one of the first mechanical calculators to help with his father's tax arithmetic.

1833 — ANALYTICAL ENGINE

Charles Babbage designs a general-purpose machine with memory, a "mill" and punched-card input — a computer in everything but electronics.

1843 — ADA LOVELACE

Her Notes contain what is widely regarded as the first published computer program — an algorithm written for a machine that did not yet exist.

1946 — ENIAC

The age of electronic computing begins in a Philadelphia room full of glowing tubes.

1947 — TRANSISTORS

Bell Labs replaces the vacuum tube: smaller, cooler, far more reliable.

1958 — INTEGRATED CIRCUITS

Whole circuits etched onto single chips. Computing begins to shrink.

1971 — MICROPROCESSOR

The Intel 4004 puts an entire processor on one chip. Computing becomes smaller, faster, personal.

1977-81 — PERSONAL COMPUTER

Machines arrive on desks and in homes. Computing becomes ours.

1990s — INTERNET

The web opens to everyone. Computers become globally connected.

2000s — BIG DATA

The world begins generating information at unprecedented scale.

2010s — MACHINE LEARNING

Instead of following hand-written rules, computers learn patterns from data.

2012 — DEEP LEARNING

Neural networks reach critical scale — a deep network sweeps the major image-recognition benchmark, and the field ignites.

2022— — GENERATIVE AI

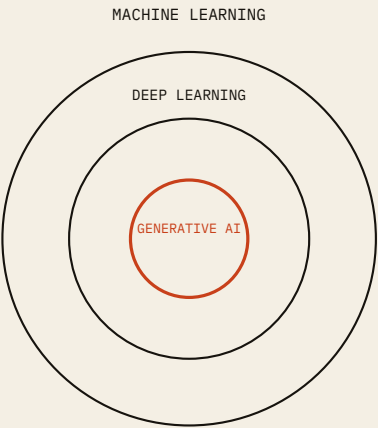
Machines generate text, images, code and audio. The journey turns from calculating to creating.

□ NOTE THE DENSITY — ACCELERATION IS THE POINT

COMPUTER + HUMAN

The computer was built to calculate. AI is being built to help us decide, create and solve. The future isn't computer versus human — it's computer plus human.

CH. 02 — THREE LEVELS OF AI



LEVEL 01

Machine Learning

Learning from data instead of following hand-written rules.

LEVEL 02

Deep Learning

Learning through layered neural networks — pattern recognition at scale.

LEVEL 03

Generative AI

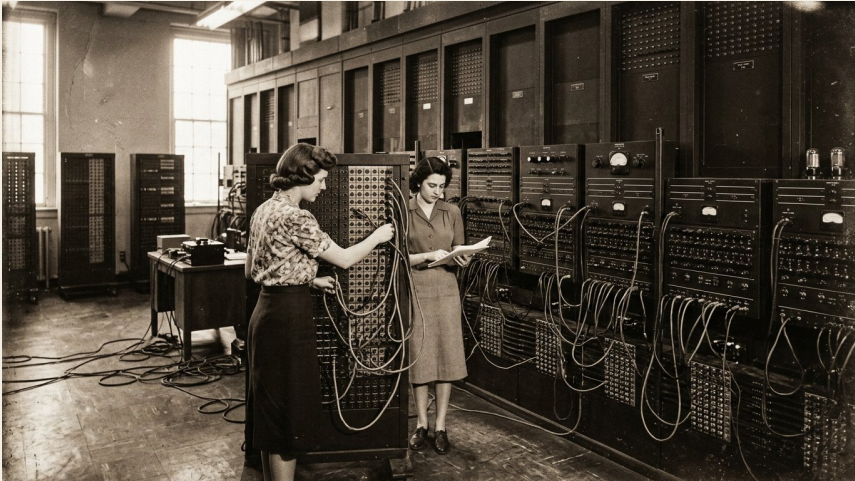
Creating new text, images, code and audio from learned patterns.

ENIAC

STORY 02 / ISSUE 01 / 1946 — THE COLLECTOR THAT STARTED THE DIGITAL AGE / 9 MIN / DOC. EN-02

ELECTRONIC NUMERICAL INTEGRATOR AND COMPUTER — ANNOUNCED FEBRUARY 1946 — MOORE SCHOOL, PHILADELPHIA

Before smartphones, laptops and AI, there was a 30-ton machine that filled a room.



CH. 01 — THE NUMBERS / SCALE AS EVIDENCE

30 TONS

DATA POINT 01 / 05 — Total weight — a machine the size of a large room.

17,468 VACUUM TUBES

DATA POINT 02 / 05 — The glowing heart of the machine.

≈1,800 SQ. FT.

DATA POINT 03 / 05 — It occupied an entire hall.

150 KW

DATA POINT 04 / 05 — Its enormous power draw.

5,000 ADDITIONS / SEC

DATA POINT 05 / 05 — Extraordinary speed for its era.

CH. 02 — WHY WAS IT BUILT?

War needed speed.

During the Second World War, the U.S. Army needed thousands of complex artillery-trajectory calculations. Human computers — working by hand — were simply too slow. So engineers built a machine that could calculate electronically. It was designed for one urgent job: out-compute the war.



NO ~~KEYBOARD.~~

NO ~~MOUSE.~~

NO ~~SCREEN.~~

Programming meant connecting cables, switches and panels by hand. One calculation could take days to set up — the program was the wiring.

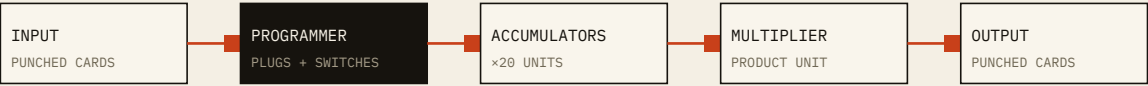


FIG. 03 — ENIAC, AS A FLOW OF SIGNALS — SOURCE: GENERAL ARCHITECTURE, SIMPLIFIED

CH. 05 — THE ENIAC SIX

Six women turned the hardware into a programmable machine.

ENIAC shipped with no manual and no programming language. Six women mathematicians — recruited from the wartime corps of human “computers” — studied its circuits and invented the craft of programming it, wiring the first ballistics runs. For decades their work went largely uncredited. History has since corrected the record.



PROGRAMMER 01 / 06

Kay McNulty

PROGRAMMER 02 / 06

Betty Jennings

PROGRAMMER 03 / 06

Betty Holberton

PROGRAMMER 04 / 06

Marlyn Wescoff

PROGRAMMER 05 / 06

Fran Bilas

PROGRAMMER 06 / 06

Ruth Lichterman

CH. 06 — WHAT CAME NEXT

ENIAC helped prove that large-scale electronic computing was possible. Its descendants are everything that followed: STORED-PROGRAM COMPUTING → MODERN COMPUTER ARCHITECTURE → THE COMPUTER INDUSTRY → TODAY’S DIGITAL WORLD

POWERED DOWN — OCTOBER 2, 1955

ENIAC was switched off. **Its idea never was.**

Every laptop, smartphone and AI system carries a piece of that revolution. One machine. One room. The start of everything since.

BCA is not just a degree.

It's a launchpad.

A degree gives you knowledge. The Bachelor of Computer Applications gives you the opportunity to build with it. You don't just study technology — you create it.

CH. 01 — YOU LEARN TO...

CODE

// TURN IDEAS INTO SOFTWARE

BUILD

// WEBSITES, APPS, DIGITAL PRODUCTS

SOLVE

// BREAK COMPLEX PROBLEMS INTO SMALLER ONES

THINK

// LOGIC AND ANALYSIS, AS A HABIT



CH. 02 — ONE DEGREE. MANY RUNWAYS.

RWY 01 THE CAREER RUNWAY

Software Developer, Web Developer, App Developer, Data Analyst, Cybersecurity Analyst, UI/UX Designer, Cloud Engineer

RWY 02 THE HIGHER STUDIES RUNWAY

MCA, M.Sc. IT, MBA / IT Management, MS & International Studies

RWY 03 THE CREATOR RUNWAY

Freelancing, Startups, Apps & Products, Digital Agencies, Entrepreneurship

CH. 03 — FIELD NOTES / WHAT THE DEGREE ALSO TEACHES

□ LOGIC

How to break problems down until they can be solved.

□ PATIENCE

Because code rarely works perfectly the first time.

□ TEAMWORK

Because great software is almost never built alone.

□ ADAPTABILITY

Because the technology itself never stops changing.

THE REAL ADVANTAGE

AI. Data science. Cloud. Cybersecurity. Automation.
The technology keeps changing — strong
fundamentals keep you ready.

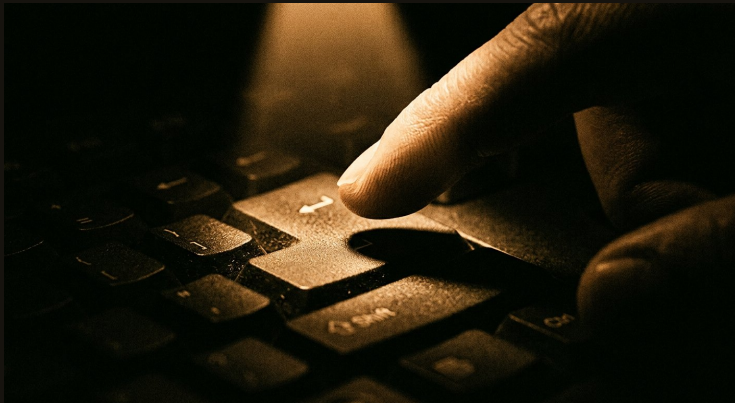
BCA IS NOT THE DESTINATION.
IT'S YOUR TAKE-OFF POINT.



CLEAR FOR TAKE-OFF – RWY 03

Your one click can cost you everything.

*In the physical world, we don't open our door to strangers.
Online? We sometimes do it with one click.*



**ONE CLICK.
ONE MISTAKE.
ONE HUGE LOSS.**

CH. 01 — INTERCEPTED / THE BAIT CHANGES, THE GOAL DOESN'T

INTERCEPT 01 — FAKE JOB OFFER

VERDICT: BAIT / SOCIAL ENGINEERING

INTERCEPT 02 — PHISHING MESSAGE

VERDICT: BAIT / SOCIAL ENGINEERING

INTERCEPT 03 — FAKE GIVEAWAY

VERDICT: BAIT / SOCIAL ENGINEERING

CH. 02 — HACKERS TARGET HUMAN EMOTIONS

FEAR. > "Your account will be closed!"

GREED. > "You've won a lottery!"

CURIOSITY. > "Is this you in this video?"

The exploit is rarely clever code. It is you, feeling something — and clicking anyway.

CH. 03 — YOUR 5 GOLDEN RULES

RULE 01 STOP. THINK. CLICK.

Check the real website before opening any link. Urgency is a red flag, not a reason.

RULE 02 PROTECT YOUR OTP.

Your OTP is a digital key. No bank, no service, no friend ever needs it. Never share it.

RULE 03 QUESTION “FREE”.

Free downloads, cracked software and too-good offers can hide malware. If it looks free, look twice.

RULE 04 USE TWO LOCKS.

Enable two-factor authentication (2FA) wherever possible. One password is a door; two is a gate.

RULE 05 UPDATE.

Software updates are not nagging – they are patches. They close the holes attackers already know about.

CH. 04 – REMEMBER

Passwords can be changed.

Money can sometimes be recovered.

But  may never be restored.

BE SMART. NOT JUST DIGITAL.

**STOP.
THINK.
CLICK.**

Young generation @ AI.

Boon, bane or both? Our parents grew up with Google. We are growing up with AI.

We don't just search anymore — we ask. And AI answers. This is the first generation that has grown up with a machine that talks back. The question is what that does to us.



CH. 01 — AI = A NEW SUPERPOWER

+ SUPER-TUTOR Learn concepts, ask questions and study at your own pace — a patient teacher that never sleeps.

+ SUPER-CREATOR Write. Design. Code. Create. Present. AI drafts, you direct.

+ SUPER-ACCELERATOR Turn an idea into a first draft in minutes instead of days.

+ NEW CAREERS Whole new roles, industries and opportunities are being created around it.

BUT EVERY SUPERPOWER HAS A PRICE

The biggest risk? Stopping ourselves from thinking.

If AI does every assignment, writes every line of code and answers every question — what happens to our own ability to think?

BOON — AI AS SUPERPOWER

Use it. Don't depend on it.

01 Ask AI to explain.
Then understand it yourself.

02 Ask AI for ideas.
Then create your own.

03 Let AI accelerate you.
Don't let it replace your ability to think.

BANE — AI AS DEPENDENCY

Three digital dangers.

01 DEPENDENCE
Using AI to avoid learning.

02 COMPARISON
AI-generated perfection can distort.

03 DECEPTION
Deepfakes and scams harder to detect.

ARE WE USING AI — or is AI using us?

CH. 03 — KEEP YOUR HUMAN EDGE

Five skills technology cannot simply download:

CRITICAL THINKING

CANNOT BE DOWNLOADED

CREATIVITY

CANNOT BE DOWNLOADED

COMMUNICATION

CANNOT BE DOWNLOADED

EMPATHY

CANNOT BE DOWNLOADED

LEADERSHIP

CANNOT BE DOWNLOADED

The future belongs to people who know how to work with AI.

Not people who blindly work for it.

Who is controlling whom?

You think you control the game. But every reward, sound, level and victory is designed to keep your brain engaged.

PLAYER → CONTROLS → GAME?

BRAIN — GAME

GAME → DESIGN → BRAIN?



CH. 01 — THE DOPAMINE LOOP

THE LOOP CLOSING ITSELF



CH. 02 — THE SAME CONTROLLER, TWO DIRECTIONS

THE GOOD — GAMES ARE NOT AUTOMATICALLY BAD

What the right games build

+ FASTER REACTIONS

Quick decisions under pressure.

+ PROBLEM SOLVING

Strategy, logic and planning.

+ TEAMWORK

Communication toward shared goal.

+ CREATIVITY

Building, designing, experimenting.

THE DARK SIDE — WHEN GAMING TAKES CONTROL

What excess takes

- ATTENTION PROBLEMS

Constant stimulation makes slower tasks harder.

- MOOD & ANGER

Excessive competitive play wears emotional control.

- SLEEP LOSS

Late-night screen time dismantles healthy sleep.

CH. 03 — THE GAMER'S GOLDEN RULES

RULE 01 THE 60-10 RULE

Play for about an hour, then give your brain a real ten-minute break.

RULE 02 NO GAMES BEFORE BED

Give your brain time to wind down. Sleep is where the day gets saved.

CHOOSE YOUR GAMES

Strategy, puzzle and creative games challenge your brain differently from purely repetitive play.

THE GAME IS NOT THE ENEMY. LOSING CONTROL IS.

Ask yourself: am I playing the game — or is the game playing me?

YOUR BRAIN IS THE MOST POWERFUL GAMING SYSTEM YOU OWN. PROTECT IT.

Colophon.

Technology. People. Ideas.

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STORY 03 — BCA AND COMPUTING EDUCATION

STORY 04 — CYBERSECURITY

STORY 05 — YOUNG GENERATION AND AI

STORY 06 — HUMAN BRAIN AND COMPUTER GAMES

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**We built machines to think
faster.**
**Now we must learn how to think
better.**